UNIVERSITY OF CALIFORNIA,
SANTA CRUZ

Increasing Viability of Turbine-Based Combined Cycle Jet Engines for Commercial Aviation

a literature review submitted in partial satisfaction of the requirements for the degree of

BACHELOR OF SCIENCE

in Physics (Astrophysics)

By

Zachary Warner

17 March 2025

Abstract

With recent aviation research possibly opening the door for supersonic (faster than the speed of sound) commercial flight, novel engine designs which can better take advantage of an uncapped speed limit are needed. The predominant design is the turbine-based combined cycle (TBCC) engine, which pairs a turbofan engine for low speeds, like the ones used in almost all jet aircraft today, with a ramjet engine, which is optimized for higher speeds. As the turbofan reaches its top speed, the engine switches to ramjet power to continue accelerating. These engines, however, suffer from several key problems, like poor turbofan performance above high-supersonic speeds, insufficient ramjet thrust at and below the turbofan's top speed, and inefficiencies in the mechanical engine transition process, which make the mode transition not reliable enough for the high standards of commercial aviation. Over the course of this literature review, I explore several solutions to these problems that are being actively researched.

The first of these methods is using chemical reactions to pre-cool the air before entering the turbofan, increasing its top speed and making it easier for the ramjet to start properly. This was verified by a research group's model, demonstrating an ideal increase in top speed from Mach 3 to Mach 4.91.[4] Another method was the optimization of the ramjet nozzle expansion surface, which a group of researchers found increases the thrust coefficient by 0.8% from Mach 1.5 to 3.5.[6] I also reviewed a study that used a Computational Fluid Dynamics model to study the effects of redesigning the ramjet inlet to create less drag-inducing shockwaves, which found this increased ramjet efficiency from 23.28% to 46.52%.[8] Finally, I reviewed methods to optimize the mechanical transition process, such as using a variable geometry ramjet combustor and mathematically calculating an ideal starting speed window for the mode transition process, which resulted in an optimal range of Mach 2.85 to 3.2, though this could be calculated more

accurately for the specific engines of each individual aircraft.[9] With these modifications, I believe high-supersonic and even hypersonic commercial flight is within our grasp, which would begin a revolution in shipping, logistics, and even everyday travel.

Table of Contents

Abstract 1

List of Figures 4

Acknowledgements 5

1 Introduction 6

2 Increasing Turbofan Efficiency with Chemical Precooling 9

3 Methods for Improving Ramjet Efficiency at Mode Transition Speeds 12

4 Optimizing the Mode Transition Process 16

5 Conclusion 19

References 22

List of Figures

1 Diagram of chemically precooled turbofan 11

2 Cross-section of TBCC engine nozzle 13

3 Comparison of open inlet to cone inlet 14

4 Specific impulses and thrusts of turbofan and ramjet engines 17

Acknowledgements

I would like to thank Professor Aiming Yan for her useful feedback and help with the editing process. I would also like to thank Auden Chen and Karthik Muthukumar from my group meetings for their helpful comments and suggestions.

Chapter 1: Introduction

Currently, commercial jets all travel at subsonic speeds (less than the speed of sound) because sonic booms caused by exceeding the sound barrier would be a constant nuisance to people on the ground below. However, in the near future, this may not be an issue, with experimental aircraft like NASA's X-59 showing promising results in greatly dampening the volume of a sonic boom from the ground, opening the door for higher speeds in commercial aviation. Without having to worry about sonic booms, commercial jets with modern technology could easily reach supersonic speeds (greater than the speed of sound) around Mach 2-3 (two to three times the speed of sound), as military jets already do.

This increased speed of travel would allow more commercial operations, like travel and shipping, to be completed in the same amount of time, in theory, lowering the costs. However, in practice, the time saved by flight at Mach 2-3 does not outweigh the cost of the loss in fuel efficiency currently associated with supersonic flight. If the aircraft could be made to fly faster and thus save more time though, the time saved could be significant enough to warrant a shift in the world of commercial aviation. This is what is motivating the push for hypersonic flight, flight at speeds where the incoming airflow chemically reacts upon contact, causing air molecules to vibrate or even break down, and heat transfer between the air and the aircraft means that special materials must be used in order for the aircrafts leading edges not to melt.[1] However, to achieve this, commercial jets will require a novel type of engine that can sustain flight at these speeds.

The world of commercial aviation currently runs on the back of the turbofan engine, the engine that powers almost every passenger, cargo, and military jet in service today. This type of engine works by using a series of turbines to pull air into the engine inlet and compress it to a high pressure as it reaches the burner, where it is mixed with fuel and ignited, passing through

[1] "Hypersonic Aerodynamics," NASA, https://www.grc.nasa.gov/www/BGH/index.html

some more turbines and the nozzle with much more energy, providing thrust for the aircraft.[2] These engines work very well for our purposes and can be made to be very efficient, however, due to their design, they have an effective speed limit, as once they get too fast, the air hitting the turbine blades causes too much drag to produce any more net thrust and could cause shockwaves in the engine, which could damage it. For turbofan engines designed for the highest possible speeds, this limit appears at Mach 2-3.

One method of propelling an aircraft beyond this limit, even up to hypersonic speed, is through the use of ramjet or scramjet engines, jet engines which have no moving parts to suck air through the engine like a turbofan, and instead rely on the movement of the aircraft to push air into the engine at high speeds, where it is mixed thoroughly with fuel and ignited just before a nozzle to provide thrust[3]. Scramjet engines only differ in that they do not slow the incoming air to subsonic speeds before this process. Since these engines have no moving parts like a traditional turbine engine for the air to collide with at supersonic speeds, they operate very efficiently at supersonic and hypersonic speeds (usually greater than Mach 3). However, without turbine blades to suck in air on their own, ramjets and scramjets do not work for takeoffs and landings and have little to no efficiency at low supersonic speeds and below.

In order to get around this problem, a ramjet or scramjet powered aircraft must first be brought up to supersonic speeds by some other means of propulsion. As of now, the main way nations around the world are attempting to achieve this for reusable aircraft is by pairing the ramjet or scramjet engine with a conventional turbofan engine used for low speeds, which the airflow is then diverted away from when the aircraft is at suitable speeds for the ramjet or scramjet. This pairing is called a turbine-based combined cycle (TBCC) engine, and while it may

[2] "Turbofan Engine," NASA, https://www.grc.nasa.gov/www/k-12/airplane/aturbf.html
[3] "Ramjet Propulsion," NASA, https://www.grc.nasa.gov/www/k-12/airplane/ramjet.html

be the most reliable option, it still has several issues hindering its use in commercial aircraft, and minimizing these issues is what I will focus on in this literature review.

The major problems faced by TBCC engines are an inefficient transition between turbofan and ramjet power, poor turbofan efficiency at higher supersonic speeds where the ramjet is not efficient enough yet to fly on its own, and poor ramjet efficiency at speeds where the turbofan is no longer efficient enough. The common factor with these problems is that the efficiency regimes of the turbofan and ramjet engines overlap only slightly where both of their efficiencies are not ideal, leading to inefficient or even failed transitions between turbofan and ramjet modes, making combined cycle engines currently unsuitable for commercial aviation.

Over the course of this literature review, I will investigate several possible solutions to these problems, including using chemical pre-cooling to increase the efficiency and top speed of the turbofan engine, optimization of the nozzle and inlet to increase ramjet efficiency at lower supersonic speeds, and optimizing how the mode transition process is controlled to maintain proper airflow during the transition. I will first review the use of chemical precooling to increase turbofan efficiency at higher speeds, allowing for the ramjet to take over much more consistently. Then, I will review methods aimed at increasing ramjet efficiency at lower speeds, like optimization of the nozzle expansion surface and a redesign of the inlet to reduce drag caused by shockwaves. Finally, I will explain how these methods can be used in conjunction with an optimized transition process to provide a large overall improvement to the efficacy of turbine-based combined cycle engines and discuss the future of where combined cycle engines could go and what improvements could possibly be made in the future.

Chapter 2: Increasing Turbofan Efficiency with Chemical Precooling

A large part of the problems facing the commercial application of turbine-based combined cycle engines is the significant drop in efficiency of the turbofan engine as the aircraft reaches speeds greater than Mach 2.5, which makes it less likely that the ramjet will be able to take over in the mode transition.[4] The important factors in determining the efficiency of a jet engine are the speed of the aircraft (and thus the speed of the incoming air) and the speed at which the exhaust leaves the nozzle at. One of the key factors that determines the effective speed limit of turbofan engines is the net thrust, which is the force of the exhaust exiting the nozzle minus the force of the air entering the engine.[5] These forces are the products of the mass flow rate of the air/exhaust and its speed upon entering/exiting the engine, and for high performance jet engines, the exhaust exit speed is maximized for greater speeds, which in turn increases the incoming air speed.[5]

A possible way to increase the net thrust is currently being explored in placing a heat exchanger in front of the turbofan's compressor fan, significantly lowering the inlet temperature of the compressor, which allows the engine to accelerate the exhaust to higher speeds. A heat exchanger is simply a series of tubes containing fluids with high heat absorption capacities to take heat from the air passing around the tubes. Different substances have been tested for use in the heat exchanger in different engine designs, including the aircraft's fuel, liquid hydrogen, liquid helium, and combinations of the three in multi-stage heat exchangers. Designs like the SABRE engine from England, however, found that using fuel as coolant at Mach 5 caused a 2.8 times increase in fuel consumption from just burning the fuel.[4] While the SABRE engine is a bit

---

[4] Wang, Cong, et al. "Thermodynamic Analysis of Chemical Precooled Turbine Combined Engine Cycle." Energy Conversion and Management, https://doi.org/10.1016/j.enconman.2021.114184.

[5] Benson, Tom. "General Thrust Equation." NASA, www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/thrsteq.html#:~:text=Across%20the%20exit%20area%20we,airflow%20will%20produce%20high%20thrust.

of a different design, utilizing a rocket engine with a turbo-compressor, the precooling concept for the compressor is essentially identical, so TBCC engines experience the same increase in fuel consumption if fuel is used as the coolant. This fuel consumption increase happens because the fuel undergoes a chemical reaction when heated significantly by the incoming air, so some of the fuel used in the precooler becomes unusable and must be dumped. Of these options, the scientific community seems to agree that liquid hydrogen works the best, due to its high heat absorption capacity.[4]

All of the precooling methods listed so far rely purely on physical interactions, however, utilizing a chemical reaction called thermal cracking in the precooler can actually increase the heat absorption capacity by up to 1.5 times.[4] This process uses the heat from the incoming air to break down long hydrocarbon chains into shorter chains, converting thermal energy into chemical energy. This works perfectly for TBCC engines which run on hydrocarbon fuels, as the higher heat absorption capacity due to the reaction helps to reduce the increase in fuel consumption from using fuel as coolant. For the temperature range of incoming air encountered at Mach 2.5-4 (where increased efficiency is most important for the mode transition), alcohol fuels like methanol (CH_3OH) and ethanol (C_2H_5OH) have the desired cracking temperature, meaning their heat absorption capacities can be boosted significantly by the reaction.[4] While alcohol fuels are excellent for the precooling process, traditional hydrocarbon fuels like aviation kerosene (RP-3) are still more efficient for the combustion process, so the best current solution is to use a dual-fuel system with methanol and RP-3 to capitalize on the benefits of both and maximize efficiency. Below is a diagram of the processes happening in a chemically precooled turbofan as it operates (see Figure 1).

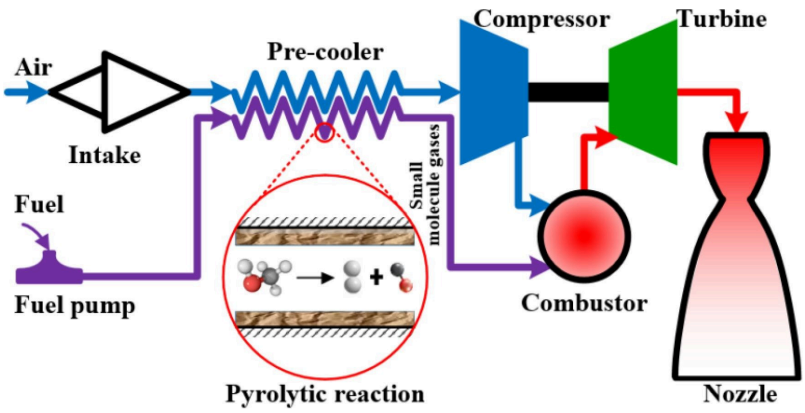


Figure 1: Diagram of the key processes involved in a chemically precooled turbofan engine in operation, emphasizing the pyrolytic reaction, thermal cracking, happening in the precooler to absorb heat from the incoming air and break down the fuel into smaller hydrocarbons.[4]

With a thermodynamic model of a TBCC engine with a one to one ratio of methanol and RP-3 and under ideal conditions, researchers from the Harbin Institute of Technology simulated an increase in the highest operational mach number (the speed at which the net thrust of the ramjet overtakes that of the turbofan) from Mach 3 without precooling to Mach 4.91 with optimal precooling.[4] They also found that utilizing a dual-fuel system with methanol and RP-3 reduced the fuel consumption from just using RP-3 as coolant by 40.97% to 50.11% at speeds of Mach 4-5, proving that methanol cracking has a massive effect on fuel efficiency.[4] Even if the lower end of these top speeds at Mach 4 is achievable in an actual aircraft as part of a turbine-based combined cycle engine, this would already make the mode transition to ramjet power much more consistently achievable and efficient, though there are still other methods that could be used to increase the consistency even more.

Chapter 3: Methods for Improving Ramjet Efficiency at Mode Transition Speeds

With chemical precooling allowing for reliable turbofan power up to possibly Mach 4, an efficient mode transition to ramjet or scramjet power would already be more likely. However, for commercial aviation purposes, the mode transition must be practically flawless and consistently replicable. Another step the aviation industry could take toward this would be to improve the efficiency of the ramjet or scramjet part of the turbine-based combined cycle engine at mode transition speeds, which, based on the chemically precooled turbofan top speed of Mach 4, would take place in the range of Mach 3-4. Possible methods for increasing ramjet/scramjet efficiency include the aerodynamic optimization of the nozzle and designing the inlet to create oblique shocks in the incoming air instead of normal shocks.

At these mode transition speeds and beyond, a large factor limiting the efficiency of ramjet and scramjet engines is the loss of thrust caused by complex shockwave interactions in the engine due to supersonic air entering the engine.[6] Scientists attempted to mitigate this problem in 2018 by optimizing the shape of the ramjet/scramjet nozzle, using a Reynolds Averaged Navier-Stokes (RANS) solver to model the shock fronts. RANS equations are used, as they describe how the working fluid, air, flows through the engine over time, allowing them to determine the speed of the exhaust and therefore the thrust. They used this model of the shock fronts to test iterative design changes by a differential evolution algorithm, specifically to the shape of the ramjet nozzle expansion surface, where the exhaust leaving the ramjet expands against to increase exhaust pressure and speed, which equates to more thrust (see Figure 1).

[6] Li, Shangze, et al. "Aerodynamic Optimization of Turbine Based Combined Cycle Nozzle." International Journal of Turbo & Jet - Engines, https://doi.org/10.1515/tjj-2016-0052.

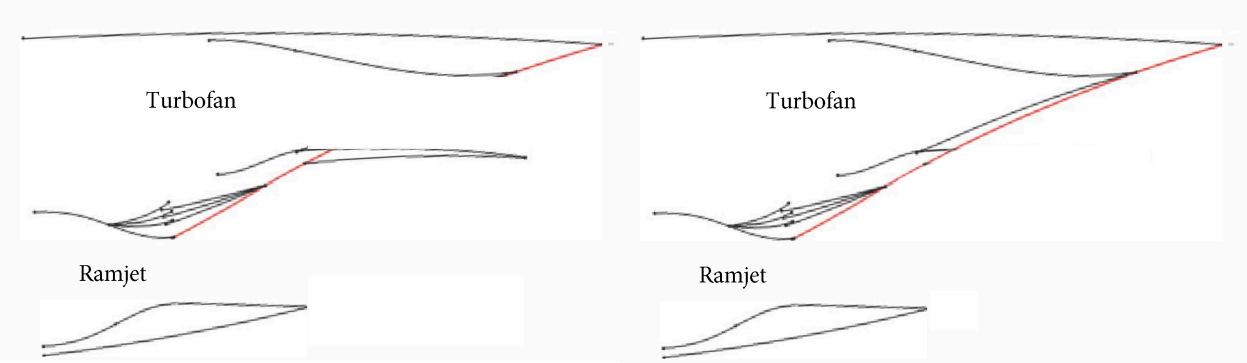


Figure 2: Cross section of a combined cycle engine's nozzle section (solid surfaces marked in black) with the exhaust diverter panel shown in both the open position (left), allowing turbofan exhaust out of the nozzle, and in the closed position (right), blocking the turbofan exhaust path and becoming part of the ramjet nozzle expansion surface marked in red.[6]

The quantity they were interested in optimizing for increased ramjet efficiency was the thrust coefficient (C_T), the ratio between the actual thrust (the force pushing on the exit section of the nozzle) and ideal thrust (the maximum theoretical force based on the mass flow, temperature, and nozzle pressure ratio).[6] The iteration of the ramjet nozzle expansion surface that was found to increase the thrust coefficient the most improved it by about 0.8% from Mach 1.5 to Mach 3.5, with the improved ramjet now able to achieve a thrust coefficient of 97.4% at Mach 3.0, while it experienced a loss of only 0.1% at Mach 4.[6] This improvement, while not massive, already makes the mode transition more reliable at Mach 3 and above and would be simple to implement alongside the other ramjet/scramjet modifications I will explain in the remainder of this chapter.

Another method that has been shown to increase ramjet and scramjet efficiency is the modification of the inlet to create oblique shock waves in the incoming air, instead of normal shocks, which occur when the inlet is completely open. In these engines, these shock waves are created when the supersonic incoming air hits the inlet, and depending on the shape of the inlet, they can either be normal (traveling perpendicular to the airflow) or oblique (traveling inclined

toward the airflow).[7] With the shock wave traveling inclined to the flow, the airflow is slowed down less before hitting the combustor, leading to less drag and higher efficiency. We can change the normal shocks that form in a fully open inlet into oblique shocks rather simply by changing the geometry of the ramjet or scramjet's inlet to a cone with the tip facing forward secured to the front of the inlet. Now, instead of hitting a flat, open hole and causing normal shocks, the airflow hits the spike first, causing oblique shocks in the flow path as the air is pulled around the base of the cone and into the inlet, as shown in the figure below (see Figure 2).

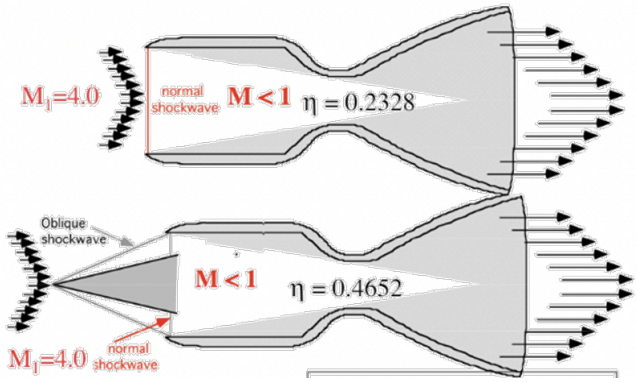


Figure 3: Standard open inlet design (top) versus inlet with spike/cone (bottom). "η" represents efficiency and "M" represents the speed of the air in Mach number. Airflow is represented by the arrows, moving left to right, and the lines labeled "shockwave" represent the path traveled by normal and oblique shockwaves.[8]

A researcher from the University of Idaho used Computational Fluid Dynamics (CFD) to model the efficiency of two inlet designs for ramjets and one for scramjets: the standard open design that creates normal shocks, a spike/cone design that creates an oblique shock first before a smaller normal shock, and a spike/cone design that creates two oblique shocks only, which would only apply to a scramjet, as the incoming air is still supersonic when it reaches the burner.[8] Computational Fluid Dynamics was vital for simulating the supersonic fluid flows, temperatures,

[7] "Oblique Shock Waves." Edited by Nancy Hall, NASA, 13 May 2021, www.grc.nasa.gov/www/k-12/airplane/oblique.html.

[8] Abbass, Amr. "A Numerical Model for Scramjet and Ramjet Efficiency Study During Dual and Individual Modes of Operation." Journal of Management and Engineering Sciences, https://doi.org/10.61552/jmes.2024.02.002.

and shockwaves in and around the engine. With the airflow, temperatures, and shockwaves modeled at all points of the engine's thermodynamic cycle, they were able to calculate several key factors of thrust performance, such as specific thrust, specific impulse, and propulsive and overall efficiencies. In the ramjet case, adding the cone/spike to the inlet increased the efficiency, by their calculations, from 23.28% to 46.52%, representing a significant increase in capability at both high and low speeds.[8] The two oblique shocks in the scramjet case resulted in a further increase of 6-10%.[8]

These benefits of the inlet redesign combined with the increased thrust coefficient gained by optimizing the ramjet nozzle expansion surface and possibly other components of the engine would allow the ramjet or scramjet part of a turbine-based combined cycle engine to reach greater efficiencies in the Mach 3-4 mode transition regime. This would bring us even closer to our goal of perfectly consistent transitions from turbofan to ramjet power, which is necessary for commercial viability. With a fairly sizeable regime of efficiency overlap between the turbofan and ramjet/scramjet engines, the consistency of mode transitions is now mostly influenced by the actual mechanical process of switching from turbofan to ramjet power, so in the next chapter, I will focus on methods researchers have explored to optimize the mode transition process.

Chapter 4: Optimizing the Mode Transition Process

Now that the turbofan and ramjet/scramjet performance envelopes overlap significantly more with these adjustments, the last remaining hurdle to commercially viable turbine-based combined cycle engines is thrust loss that occurs during the mechanical mode transition process. This thrust loss comes from aerodynamic drag on the moving parts for the mode transition, as well as a decreased amount of air entering the turbofan before the ramjet/scramjet can provide enough thrust on its own.[9] This could create a jarring motion for the aircraft and its passengers or cargo, or it could even cause the ramjet/scramjet to not start properly, so this problem must be addressed before these engines could be used in commercial aircraft.

Researchers from Harbin Institute of Technology studied this problem extensively, using a computational fluid dynamics model to test three possible methods of improving the mode transition process.[9] Of these methods, two are relevant to the type of TBCC engine I have been discussing here, those being the use of a computational fluid dynamics model to mathematically determine the appropriate start speed of the mode transition process and the addition of a variable geometry section to the ramjet or scramjet's combustor, which is adjusted during the mode transition to allow the ramjet to operate better at the slower speeds at the beginning. In the model they used to test the optimal starting speed, they used a form of turbofan precooling (like described in chapter 2) to achieve a similar increase in the turbofan's top speed, with an increase to specific thrust of about 18.3%.[9] It is worth noting that this is actually a smaller increase to thrust than the ideal precooling methods discussed in chapter 2, so the actual ideal mode transition range could be slightly different based on the performance of the turbofan actually

[9] Lv, Chengkun, et al. "Mode Transition Analysis of a Turbine-based Combined-cycle …" Energy, https://doi.org/10.1016/j.energy.2022.125324.

being used. This calculation could be repeated with the specifications of the parts of any particular TBCC engine to obtain the ideal mode transition range for that specific engine.

To find the optimal range of speeds for mode transition, the researchers used computational fluid dynamics models of both their precooled turbofan engine and ramjet engine to plot the specific impulse and specific thrust of each engine over a range of speeds from Mach 2.5-3.5. The researchers overlaid these plots and noted the speed regimes where the specific impulses and specific thrusts overlap, testing over a range of ramjet combustor expansion ratios from 1.2 to 2.5. A combustor expansion ratio in a ramjet is the ratio of the volume of the air-fuel mixture before combustion to after combustion, which affects the specific thrust and impulse differently at different speeds. The overlaid plots are shown below (see Figure 3).

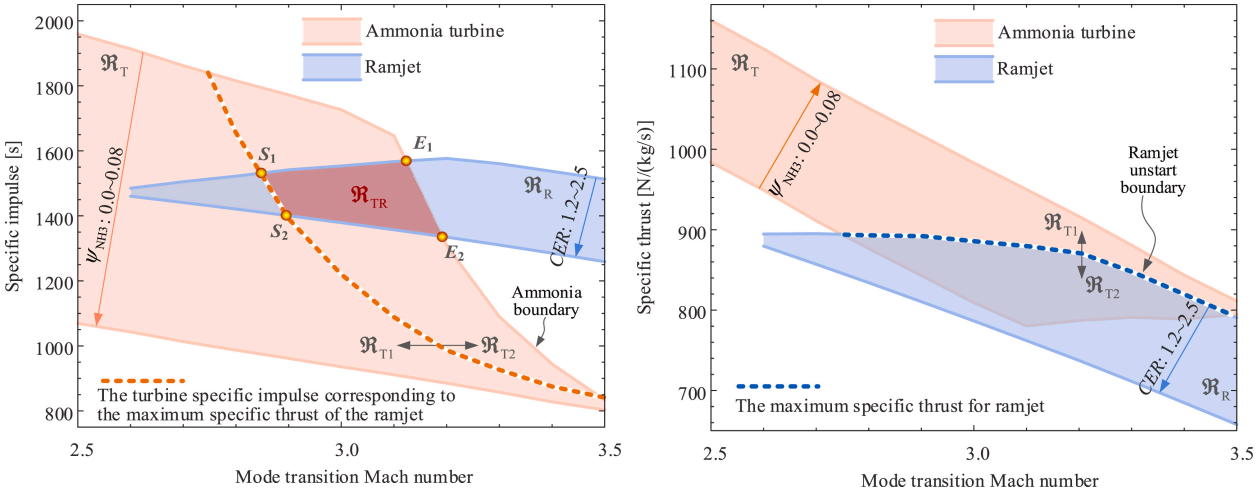


Figure 4: Specific impulse versus mode transition speed in Mach number of the precooled turbofan and ramjet (left) and specific thrust versus mode transition speed in Mach number of the precooled turbofan and ramjet (right). "CER" is the combustor expansion ratio and "Ψ_{NH3}" is the mass ratio of coolant to air, which can be adjusted on any precooler. The red highlighted area on the specific impulse graph marks the speed range where the specific thrusts and impulses of both engines overlap.[9]

With this overlap range noted, the researchers saw that the mode transition process could start somewhere in the range of Mach 2.85 to 3.2, as this is where the specific thrust and impulse of the ramjet begin to approach those of the precooled turbofan.[9] With their precooling method,

starting the mode transition below Mach 2.8 would make the transition to ramjet power fail, as the specific thrust of the ramjet would be less than that of the precooled turbofan, so the aircraft would begin to slow down further, making it impossible for the ramjet to keep providing thrust. Above Mach 3.2, their precooled turbofan provides less specific thrust and impulse than the ramjet, meaning the aircraft would not be able to go any faster without switching to the ramjet. If the precooled turbofan could provide more thrust at higher speeds however, like the researchers discussed in chapter 2 were able to achieve, this upper limit on the mode transition start speed would be greater.

This range of speeds is helpful to know when deciding the best transition point for a turbine-based combined cycle engine, however, in order for the ramjet to start at the lower end of this range, its combustor expansion ratio would have to start closer to 2.5 and decrease during the mode transition.[9] The researchers concluded that this could be achieved by adding a section to the ramjet's combustor that can move to restrict the pre-combustion volume during the mode transition process, decreasing the combustor expansion ratio as the ramjet engine begins accelerating the aircraft.[9] This variable geometry section can be incredibly simple in design, as all it needs to do is constrict the volume of the pre-combustion fuel-air mixture as the speed rises. With a variable geometry ramjet combustor, the mode transition may start at any speed within the Mach 2.85 to 3.2 range determined mathematically using a computational fluid dynamics model of the engine.

Chapter 5: Conclusion

Commercially viable high-supersonic and hypersonic flight would mean a revolution in shipping, logistics, and possibly even passenger flight. The almost universally accepted solution to reliable and safe propulsion up to sustained hypersonic speeds is the turbine-based combined cycle engine, although this technology still has several important issues that need to be addressed in order to be considered reliable enough for commercial use. These key issues (poor turbofan efficiency at high-supersonic speeds and above, insufficient ramjet thrust at and below maximum turbofan speeds, and an inefficient transition from turbofan to ramjet) can be addressed by recent experimental advances in engine design, opening the door for commercial hypersonic aviation. As discussed in this literature review, these advances include a chemical precooling stage for the turbofan engine, computational optimization of ramjet parts like the nozzle expansion surface, a redesigned ramjet inlet to reduce drag caused by normal shockwaves, a mathematically determined mode transition start speed, and a variable geometry ramjet combustor.

To solve the first problem of poor turbofan efficiency at high-supersonic speeds, I looked to a technology called chemical engine precooling, where the high speed, incoming air first passes through a heat exchanger to cool it before it reaches the turbofan's compressor, allowing for higher thrust output at higher speeds. With a one-to-one ratio of aviation kerosene (RP-3) to methanol in the heat exchanger, researchers found that, under ideal conditions, a turbofan engine's effective top speed was increased from Mach 3.0 to Mach 4.91, vastly increasing the likelihood of a successful mode transition.[4]

The second problem of insufficient ramjet thrust at the top end of turbofan speeds is remedied using two ramjet modifications being actively researched, these being the computational optimization of the nozzle expansion surface and a geometric inlet redesign.

Computational fluid dynamics simulations and models of these modifications demonstrate that they both significantly increase ramjet thrust and efficiency at lower speeds, adding to the reliability of the transition from turbofan to ramjet propulsion.[6,8]

Finally, to increase the overall efficiency of the mode transition process, the start speed of the mode transition process is an incredibly important factor. Researchers went about finding this optimal start speed by studying the speeds at which the specific thrusts and impulses of both engines overlap. They found that with an additional variable geometry section to the ramjet's combustor to allow it to start at lower speeds, there exists a region from Mach 2.85 to 3.2 where the turbofan can keep providing sufficient thrust while the ramjet starts and begins accelerating the aircraft under its own power.[9] This calculation could be done with the specifications of the engines being used on any particular aircraft to find the ideal mode transition start speed for that specific plane, increasing the consistency and reliability of the process.

Looking forward, there are still some issues with turbine-based combined cycle engines that must be addressed before they are widely adopted in the commercial world. Issues like determining the ideal materials to use in the design of the engines and aircraft to increase durability to high temperatures, as well as settling on the proper fuel mixture still need to be agreed upon.[10] There are also economic factors that currently hinder the use of turbine-based combined cycle engines, such as manufacturing, operating, and fuel costs, which must be taken into account when designing for future aircraft.

However, even with these challenges, the advancements I have highlighted in this review show that hypersonic flight using turbine-based combined cycle engines is incredibly close to viability in commercial aviation. With continued research and development on this type of

[10] Zuo, Feng-Yuan, and Sannu Mölder. "Hypersonic Wavecatcher Intakes …" Progress in Aerospace Sciences, https://doi.org/10.1016/j.paerosci.2019.03.001.

engine, a revolution in aviation as we know it is possible, where anything or anyone can be flown to any part of the world in no more than a few hours. This development has massive implications for the aviation industry and so many others, making the turbine-based combined cycle engine an incredibly important innovation to pursue, and the methods discussed in this review represent a huge step toward that goal.

References

Abbass, Amr. “A Numerical Model for Scramjet and Ramjet Efficiency Study During Dual and Individual Modes of Operation.” Journal of Management and Engineering Sciences, vol. 1, no. 2, 31 May 2024, pp. 36–48, https://doi.org/10.61552/jmes.2024.02.002.

Benson, Tom. “General Thrust Equation.” Glenn Research Center, NASA, www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/thrsteq.html#:~:text=Across%20the%20exit%20area%20we,airflow%20will%20produce%20high%20thrust.

“Hypersonic Aerodynamics”, Edited by Tom Benson, NASA, 7 May 2021, www.grc.nasa.gov/www/BGH/index.html.

Li, Shangze, et al. "Aerodynamic Optimization of Turbine Based Combined Cycle Nozzle." International Journal of Turbo & Jet - Engines, vol. 35, no. 4, 2018, pp. 385-394. ProQuest, https://www.proquest.com/scholarly-journals/aerodynamic-optimization-turbine-based-combined/docview/2174301921/se-2, doi:https://doi.org/10.1515/tjj-2016-0052.

Lv, Chengkun, et al. “Mode Transition Analysis of a Turbine-based Combined-cycle Considering Ammonia Injection Pre-Compressor Cooling and Variable-Geometry Ram-Combustor.” Energy, vol. 261, 15 Dec. 2022, https://doi.org/10.1016/j.energy.2022.125324.

“Oblique Shock Waves.” Edited by Nancy Hall, NASA, 13 May 2021, www.grc.nasa.gov/www/k-12/airplane/oblique.html.

“Ramjet Propulsion.” Edited by Nancy Hall, NASA, 13 May 2021, www.grc.nasa.gov/www/k-12/airplane/ramjet.html.

“Turbofan Engine.” Edited by Nancy Hall, NASA, 13 May 2021, www.grc.nasa.gov/www/k-12/airplane/aturbf.html.

Wang, Cong, et al. “Thermodynamic Analysis of Chemical Precooled Turbine Combined Engine Cycle.” Energy Conversion and Management, vol. 239, July 2021, pp. 114–184, https://doi.org/10.1016/j.enconman.2021.114184.

Zuo, Feng-Yuan, and Sannu Mölder. “Hypersonic Wavecatcher Intakes and Variable-Geometry Turbine Based Combined Cycle Engines.” Progress in Aerospace Sciences, vol. 106, Apr. 2019, pp. 108–144, https://doi.org/10.1016/j.paerosci.2019.03.001.